

Meta Relation-Aware Metric Learning Framework for Few-Shot Uncertain Knowledge Graph Completion

Xinyan Wang¹, Kuo Yang¹, Jian Liu¹, Xianan Li¹, Rui Hua¹, Kuan Xu¹, Zixin Shu¹, and Xuezhong Zhou¹

Abstract—Uncertainty is a fundamental aspect of human knowledge, widely present in various domains such as medicine, biology, and finance. Uncertain Knowledge Graph Completion (UKGC) tasks encompass both link prediction and confidence prediction. However, the inaccessibility of high-quality knowledge results in sparse knowledge graphs, where a large number of entities exhibit relatively sparse neighborhoods. This sparsity severely hinders the performance of few-shot UKGC models, which generally rely on neighborhood aggregation to enhance entity representations. In this study, we propose a Meta relation-aware Metric learning framework designed for few-shot UKGC, named MMUC. To mitigate neighbor sparsity, MMUC introduces a relation-centric paradigm that decouples representation enhancement from background graphs, integrating an uncertainty-driven similarity measure through a learnable confidence learner to achieve robust, non-linear fusion for joint link and confidence prediction. Extensive evaluations across multiple benchmarks demonstrate MMUC’s superior performance through rigorous link and confidence prediction, ablation studies, and sparsity analysis, highlighting its particular resilience to neighbor sparsity in few-shot uncertain KGs.

Index Terms—Uncertain knowledge graph, few-shot learning, link prediction, confidence prediction.

I. INTRODUCTION

KNOWLEDGE Graph Completion (KGC) is a fundamental task in the field of Knowledge Graphs (KGs), aiming to infer missing facts to mitigate the inherent incompleteness of KGs [1], [2], [3]. Most existing studies focus on deterministic KGs, where all observed facts are assumed to be fully trustworthy. However, with the advancement of information extraction

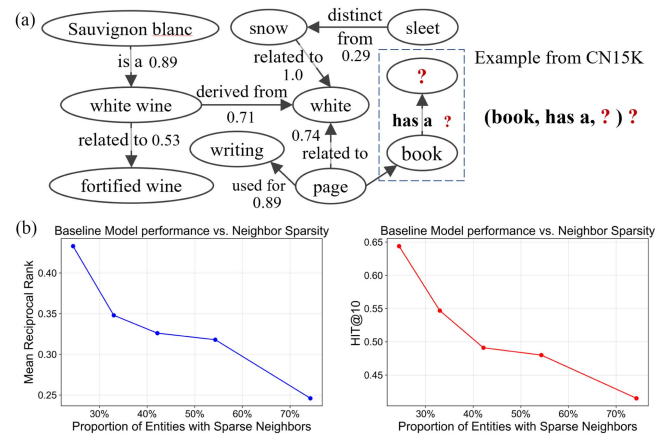


Fig. 1. (a) A UKGC example from the CN15 K dataset. (b) Performance degradation of baseline model (i.e., GMUC) under increasing entity neighbor sparsity on the NL27 K dataset.

techniques [4], KGs are increasingly constructed through automated pipelines [5], introducing inevitable noise and uncertainty into the facts. In addition, uncertainty is intrinsic to many real-world domains [6], such as medicine [7], biology [8], and finance [9]. Uncertain Knowledge Graph Completion (UKGC) thus aims not only to predict missing links but also to quantify the uncertainty of these predictions, as illustrated in Fig. 1(a). Uncertain Knowledge Graphs (UKGs) have been successfully applied in various applications [10], [11], [12], and accurate uncertainty estimation is essential for downstream real-world tasks [13], [14].

Despite recent advances in UKGC, most existing approaches still heavily depend on abundant background graph information. However, in practical KGs, many relations naturally have few-shot characteristics, where only a limited number of training triples are available [15]. These solutions are particularly sensitive to a challenge we term “neighbor sparsity”, where entities are connected to very few neighbors (see definition in Section III-A3), especially in few-shot scenarios [16]. For example, Fig. 1(b) shows the performance of a representative baseline, GMUC [17], which utilizes a metric learning framework that enhances entity representations by aggregating neighbor embeddings from the background graph. The results indicate that the model’s performance declines significantly as the proportion of sparse entities increases, as observed in the NL27 K dataset. To further illustrate the practical impact of neighbor sparsity, Fig. 2 presents a case study comparing specific entities with sparse and dense neighborhoods. As shown in Fig. 2(c), the

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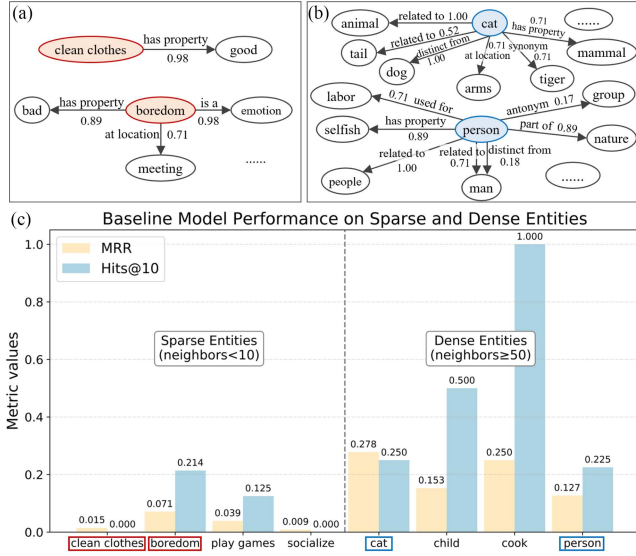


Fig. 2. (a) Local neighborhood visualization of sparse entities (“clean clothes” and “boredom”). (b) Local neighborhood visualization of dense entities (“cat” and “person”). (c) Baseline GMUC performance on sparse and dense entities, measured by MRR and Hits@10. Performance on sparse entities is consistently and significantly lower than on dense entities.

GMUC model consistently performs much worse on sparse entities (e.g., “clean clothes”, “boredom”) than on dense entities (e.g., “cat”, “person”). This performance disparity underscores the practical challenge caused by sparse local neighborhoods. Neighbor sparsity is a widespread phenomenon in real-world knowledge graphs: as shown in Fig. S1 (Section 1.1 of the supplemental material), 56.1% of entities in CN15 K and 89.7% in NL27 K have fewer than 10 neighbors. This observation reinforces the need for a framework that remains robust even in the absence of sufficient neighborhood information.

In this work, to tackle the problem of neighbor sparsity, we propose a new **Meta** relation-aware **Metric** learning framework for few-shot **Uncertain** knowledge graph **Completion**, named **MMUC**. Our core motivation is to enable the metric framework to perceive meta-relational facts within limited samples, shifting the paradigm from entity-centric neighborhood aggregation to relation-centric meta-learning. This design eliminates the dependency on background graph structures, making it more robust and effective in learning representations under sparse conditions. Specifically, our framework is built upon three tightly-coupled components: (i), the Relation Meta-aware Module (RMM) extends meta-learning to UKGs, learning a task-specific relation representation that serves as the foundation for both subsequent steps. This design effectively renders MMUC naturally immune to neighbor sparsity. (ii), the Confidence Metric Module (CMM) leverages this representation to compute an uncertainty-aware similarity between support and query triples, going beyond traditional semantic matching. (iii), a confidence learner then non-linearly fuses the plausibility score from RMM and the uncertainty similarity from CMM to produce well-calibrated confidence predictions. This synergistic integration is optimized jointly through a unified loss function. We evaluate MMUC on two benchmarks, NL27 K with its noisy variants [17], and

CN15K [6], where it consistently demonstrates strong performance across a wide range of settings. Moreover, MMUC demonstrates robust performance across varying levels of neighbor sparsity, where existing methods face performance drops of up to 40%, further emphasizing its superior generalization ability in real-world applications. In summary, our contributions are three-fold:

- We identify and formalize neighbor sparsity as a critical yet overlooked bottleneck in few-shot UKGC, proposing MMUC as a meta relation-aware metric framework specifically designed to mitigate this challenge.
- MMUC introduces a relation-centric paradigm that decouples representation enhancement from background graphs, enabling robust non-linear fusion for joint link and confidence prediction through an uncertainty-driven similarity measure.
- Comprehensive multi-benchmark evaluations demonstrate that MMUC achieves superior performance through rigorous link prediction, confidence prediction, ablation studies, and sparsity analysis, exhibiting remarkable robustness particularly in neighbor-sparse scenarios within few-shot uncertain KGs.

II. RELATED WORK

A. Few-Shot KGC

Research on few-shot KGC is a critical paradigm within the broader landscape of task-oriented knowledge graph reasoning [18], [19]. It has received growing attention in recent years due to the long-tail distribution of relations in real-world KGs. Existing approaches can be broadly divided into three types: metric learning-based, meta-learning-based, and other advanced architectures. Metric learning methods, such as GMatching [16], employ matching networks that utilize entity embeddings and local graph structures to learn a matching metric. A key strategy to improve the performance of this type is to enhance the entity pairs’ embedding to obtain richer information on relations [20], [21]. Meta-learning methods, such as MetaR [22], leverage relation-level meta-knowledge and gradient-based updates to accelerate model adaptation for few-shot link prediction. Subsequent studies further emphasize the extraction and utilization of relational information to enhance generalization [23], [24]. Other works within this paradigm, such as [25], employ a convolutional network to directly learn and match relational patterns from support triples for one-shot inference.

Beyond metric and meta-learning, other approaches have emerged for few-shot KGC. NP-FKGC [26] combines neural processes with normalizing flows to model complex distributions of KGC functions, enabling uncertainty estimation and handling complex relations via a path-aware GNN. SIM [27] employs a Transformer to enhance entity representations by modeling semantic interactions among neighbors, achieving strong results on few-shot benchmarks. This category also includes methods that leverage additional information or novel architectures, such as text-enhanced representations [28], self-supervised learning frameworks that exploit aliasing relations

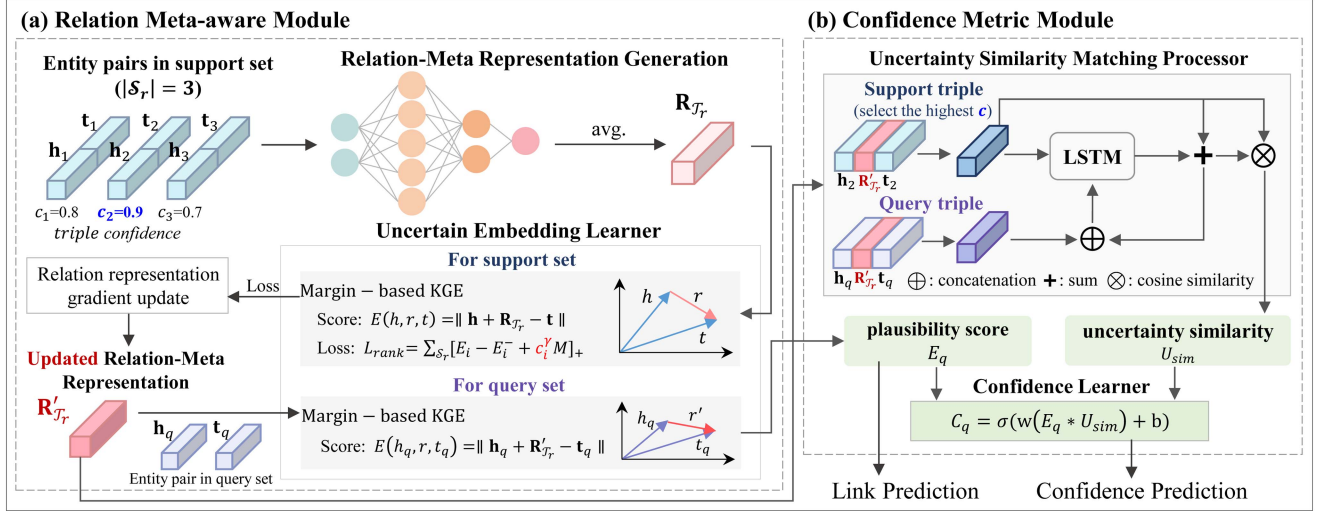


Fig. 3. An overview of the MMUC framework, which comprises two modules, (a) Relation Meta-aware Module and (b) Confidence Metric Module. $\mathbf{h}$, $\mathbf{t}$, $\mathbf{R}$ are the embedding of the head entity, tail entity, and the representation of relation r . E stands for the plausibility score and L_{LP} for margin ranking loss. c is the confidence value of triple while C is the predicted confidence. w and b are the learnable parameters, σ is the sigmoid activation function. All q refers to each triple in the query set.

[29], and frameworks aimed at low-resource KGs via multi-view task representation generation [30].

While these models are effective in deterministic few-shot scenarios, they generally lack mechanisms for handling triple confidence or uncertainty, which limits their applicability to real-world UKGs.

B. Uncertain KGC

UKGs have been proposed to model the inherent uncertainty of relational facts, which is often present in real-world scenarios. By integrating uncertainty into the KGC process, uncertain KGC methods can provide more accurate and reliable results. UKGE [6] is the pioneering method that introduces the embedding of uncertain knowledge by learning a mapping function to convert the plausibility output of KGE into confidence scores. GTransE [31] and FocusE [32] aim to capture the knowledge uncertainty by incorporating triple confidence in link prediction. However, these models assume sufficient training data and are not designed for the few-shot setting, which limits their effectiveness when relational data is scarce.

C. Few-Shot Uncertain KGC

As far as we know, GMUC [17] is the first research on few-shot learning for UKGs. It's a metric-based model that consists of a Gaussian neighbor encoder and a matching processor to capture uncertain semantic information. Building on GMUC, UFGKGC [33] introduces Gaussian-based uncertainty representations for entities and triples, and designs an uncertainty-aware relational GNN to better integrate noisy neighbor information. This framework improves robustness in few-shot settings by modeling and optimizing uncertainty during training.

Few-shot UKGC methods typically enhance entity representations through neighbor aggregation. However, this strategy is fundamentally limited by neighbor sparsity (as illustrated

in Fig. 1(b)), which severely constrains their effectiveness in real-world applications. Addressing this limitation is crucial for improving the practical applicability of few-shot UKGC. To overcome this, MMUC shifts the focus to relation-centric representations via meta-learning, directly learning from support set triples to generate relation meta-representations. By eliminating the dependence on background graph neighbors, MMUC demonstrates superior adaptability to extreme sparsity while maintaining competitive performance in both link and confidence prediction.

III. METHODOLOGY

This section presents the proposed Meta relation-aware Metric learning framework for few-shot Uncertain knowledge graph Completion (MMUC). As illustrated in Fig. 3, MMUC is designed as an integrated pipeline consisting of two synergistic modules: the Relation Meta-aware Module (RMM) and the Confidence Metric Module (CMM). The RMM first learns a task-specific relation representation from the support set through a meta-learning mechanism. This representation is then leveraged for dual purposes: (1) within RMM to score triples for link prediction, and (2) by the CMM to compute uncertainty-aware similarity between support and query triples. Finally, the CMM fuses these signals to predict confidence scores. The entire framework is optimized end-to-end by a joint loss function that couples the objectives of accurate link prediction and calibrated uncertainty estimation.

A. Preliminaries

1) *Uncertain Knowledge Graph*: An uncertain knowledge graph (UKG) is a probabilistic extension of a deterministic knowledge graph (DKG) that models the inherent uncertainty of relational facts. It's typically represented as a triple (h, r, t) with a confidence score s indicating the probability that the

fact is true, denoted as $\mathcal{G} = \{(h, r, t), c\}$ where $(h, r, t) \in \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, $\mathcal{E}$ and $\mathcal{R}$ are the entity set and relation set, respectively. Note that we follow the assumption that the confidence value $c \in [0, 1]$ and interpret it as a probability representing the degree to which the triple is true.

2) *Few-Shot Knowledge Graph Completion*: Few-shot KGC predicts KG facts with scant data, dividing triples into tasks by relation type. Each task, comprising a support set and query set sharing the same relation, requires using few support set triples to infer new query set triples. Training involves tasks $\mathcal{T}_{train} = \{\mathcal{T}_i\}_{i=1}^A$, each with support and query sets $(\mathcal{S}_i, \mathcal{Q}_i)$ sharing relation r_i , across A tasks. Testing tasks, $\mathcal{T}_{test} = \{\mathcal{T}_j\}_{j=1}^B$, involve unseen relations for fair evaluation. For UKGs, models predict both tail entities and confidence values in query sets ($\mathcal{Q}_r = \{r : ((h_j, ?), ?)\}$). Generally, the support set size, $|\mathcal{S}_r|$, is small (e.g., 1, 3, 5). Due to space constraints, we exemplify with $|\mathcal{S}_r| = 3$ for three-shot uncertain KGC.

3) *Neighbor Sparsity*: Inspired by the discussion in [34], we define the neighbor set of an entity $e \in \mathcal{E}$ in the background graph $\mathcal{G}$ as the set of all tail entities that are connected to e as a head entity:

$$\mathcal{N}(e) = \{t \mid (e, r, t) \in \mathcal{G}, \exists r \in \mathcal{R}\},$$

where (e, r, t) denotes a triple in the graph $\mathcal{G}$, with e as the head entity, r as the relation, and t as the tail entity.

The neighbor sparsity of e is then defined as the number of unique tail entities connected to e , i.e., $|\mathcal{N}(e)|$, which reflects the number of triples where e serves as the head entity. Given an average degree $\bar{d}$ in the background graph, we consider an entity as sparse if $|\mathcal{N}(e)| < \bar{d}/10$, based on the threshold used in our empirical analysis (see Section VI-A). At the graph level, the sparsity ratio is calculated as the proportion of such sparse entities. This definition allows us to quantitatively assess the impact of sparse neighborhoods on model performance.

B. Relation Meta-Aware Module (RMM)

MMUC leverages RMM to efficiently capture uncertain facts, allowing for accurate link prediction in few-shot UKGs. In addition, the plausibility scores generated from the completion model also provide a foundational basis for subsequent confidence prediction.

1) *Relation-Meta Representation Generation*: The efficacy of the relational meta-learning framework for few-shot link prediction has been demonstrated in DKGs [22]. This type of meta-learning-based method believes the most important thing for the model is to transfer the shared relation information from the support set to the query set in a specific task, and the relation information here is termed “relation-meta”. To further showcase the overall performance of our framework, the straightforward approach for generating relation-meta has been adopted in RMM. Specifically, the relation-meta representation generator is implemented as a multilayer perceptron.

For a given task $\mathcal{T}_r$, entity pair (h_i, t_i) of the triple in the support set $\mathcal{S}_r$ are the input, and a L -layer fully connected neural network is utilized to obtain the relation-meta representation

$\mathbf{R}_{\mathcal{T}_r}$. The calculation process is as follows:

$$\mathbf{x}^0 = \mathbf{h}_i \oplus \mathbf{t}_i, \quad \mathbf{x}^l = \sigma(\mathbf{W}^l \mathbf{x}^{l-1} + \mathbf{b}^l),$$

$$\mathbf{R}_{(h_i, t_i)} = \mathbf{W}^L \mathbf{x}^{L-1} + \mathbf{b}^L, \quad \mathbf{R}_{\mathcal{T}_r} = \frac{\sum_{i=1}^{|\mathcal{S}_r|} \mathbf{R}_{(h_i, t_i)}}{|\mathcal{S}_r|}, \quad (1)$$

where $\mathbf{h}_i, \mathbf{t}_i \in \mathbf{R}^d$ denote the embedding of dimension d of the head and tail entity, and $\oplus$ means concatenation. $\mathbf{W}^l$ and $\mathbf{b}^l$ are the learnable weights and bias of layer l , and $\mathbf{R}_{(h_i, t_i)}$ is the ultimate relation-meta representation for (h_i, t_i) . In cases, there are a few entity pairs in $\mathcal{S}_r$, $\mathbf{R}_{\mathcal{T}_r}$ for the current task is obtained by taking their average where $|\mathcal{S}_r|$ equals the number of triples in the support set. While mean pooling is adopted here for simplicity and stability, other pooling strategies such as max pooling, mean-max hybrid pooling, or attention-based pooling could also be considered to potentially enhance relational representation quality. We have implemented these alternative strategies and reported the corresponding results; detailed comparisons and analyses are provided in Section 4.2 of the supplemental material.

2) *Relation-Meta Representation Update*: After generating $\mathbf{R}_{\mathcal{T}_r}$, meta-learning utilizes the loss's gradient for rapid updates. We propose the uncertain embedding learner to achieve the triple scoring and gradient update. In line with Knowledge Graph Embedding (KGE) learning patterns, the most widely used TransE [35] is employed here. First, define the scoring function for each triple,

$$E(h_i, r, t_i) = \|\mathbf{h}_i + \mathbf{R}_{\mathcal{T}_r} - \mathbf{t}_i\|, \quad (2)$$

where $E(h_i, r, t_i)$ is the plausibility score of the triple (h_i, r, t_i) . Then, an uncertain generalized margin-based ranking loss considering the uncertainty of triples in the support set for UKGs can be computed as:

$$L_{LP}(\mathcal{S}_r) = \sum_{\mathcal{S}_r} [E(h_i, r, t_i) - E(h_i, r, t'_i) + c_i^\gamma M]_+, \quad (3)$$

where $[\mathbf{x}]_+$ represents the positive part of $\mathbf{x}$, (h_i, r, t'_i) is the corrupted triple for $(h_i, r, t_i) \in \mathcal{S}_r$ as the negative sample which has not appeared in $\mathcal{G}$. M is a hyperparameter for margin, and hyperparameter $\gamma \geq 0$ indicates the influence of triple confidence. When $\gamma = 0$, the formula degenerates into the margin ranking loss of deterministic KGE, which regards all triple confidence as the same 1. Larger values of γ mean a greater influence of triple uncertainty [31]. In UKGs, we set $\gamma > 0$, which means that triples with different confidences have different margins on model parameter updates. We use this generalized loss to calculate the gradient and make a rapid update of $\mathbf{R}_{\mathcal{T}_r}$:

$$\text{GD}(\mathbf{R}_{\mathcal{T}_r}) = \nabla_{\mathbf{R}_{\mathcal{T}_r}} L_{LP}(\mathcal{S}_r),$$

$$\mathbf{R}'_{\mathcal{T}_r} = \mathbf{R}_{\mathcal{T}_r} - \delta \cdot \text{GD}(\mathbf{R}_{\mathcal{T}_r}). \quad (4)$$

δ is the step size of the gradient update.

3) *Link Prediction for Triples in the Query Set*: The updated relation-meta representation is equally adopted as the relation embedding to score triples in the query set. Ranking can be performed according to the triple's plausibility score to complete the link prediction [2]. The losses of query triples are then

calculated using the (3). Finally, all losses of query sets are summed to form a part of MMUC's joint loss, which employs updating the parameters of the whole framework.

It is worth noting that the plausibility score can be employed directly to predict the triple confidence in UKGs [6]. However, Zhang's work [17] indicated that relying solely on plausibility scores for confidence prediction is inadequate in few-shot scenarios. This is reasonable because the primary purpose of the plausibility score is to facilitate the ranking of tail entities for link predictions, and the score range can vary depending on the specific KGE model. In contrast, the confidence ranges from 0 to 1 and can be interpreted as the probability of a triple being true. Unlike the plausibility scores, which focus on the relative comparison between different triples, the goal of the confidence is to measure the probability that a candidate triple is the ground truth. Therefore, in scenarios with insufficient triples, solely relying on the plausibility score to model the confidence value proves inadequate. MMUC utilizes the confidence metric module to quantify the uncertainty similarity between support and query triples. It integrates the plausibility score and maximizes the utilization of observed few samples to achieve accurate confidence prediction.

C. Confidence Metric Module (CMM)

One of the challenges in UKGC is how to predict the confidence value of a triple. A straightforward strategy is to fit the plausibility score of the scoring function modeled by KGE to the confidence value. This approach seems to be feasible but requires substantial training data to capture the appropriate correlation, making it unsuitable for few-shot scenarios. To overcome this challenge, we introduce CMM to accurately and efficiently predict the confidence of triples. The module is composed of two parts: an uncertainty similarity matching processor and a simple yet efficient confidence learner.

1) *Uncertainty Similarity Matching*: The few-shot link prediction methods based on metric learning in deterministic KGs ultimately input entity pairs into the matching processor to obtain the similarity of support and query instances and complete the link prediction according to this similarity [16]. Inspired by this, we believe that similarity matching is not only about the semantics of relational facts but also about their uncertainty. Moreover, if two triples have a high similarity, but one of them has high uncertainty, we have reason to question the confidence of the other triple as well. Therefore, our proposed uncertainty similarity matching processor aims to capture uncertainty similarities between triples from the support set and the query set.

We take inspiration from [16] and employ an LSTM-based recurrent processing block for multi-step matching. Our approach differs from the original matching processor in that we aim to capture uncertainty similarity between facts, instead of focusing solely on the similarity of entity pairs. To achieve this, we use triple representations as input to the matching processor instead of entity pairs to crystallize relation semantics. For the representation of triples, we adopt the most intuitive form of embedding addition, that is

$$\mathbf{p} = \mathbf{h} + \mathbf{R}'_{\mathcal{T}_r} + \mathbf{t}, \quad (5)$$

where $\mathbf{p}$ is the representation of (h, r, t) , updated relation-meta $\mathbf{R}'_{\mathcal{T}_r}$ from RMM is as relation embedding.

After obtaining representations of triples, we use a matching processor to compute the uncertainty similarity between the triple in the support set and the triples in the query set. The calculation process of the uncertainty similarity is as follows:

$$\begin{aligned} \mathbf{hs}'_{k+1}, \mathbf{cs}_{k+1} &= \mathbf{f}_L(\mathbf{p}_q, [\mathbf{hs}_k \oplus \mathbf{p}_{s_{top}}, \mathbf{cs}_k]), \\ \mathbf{hs}_{k+1} &= \mathbf{hs}'_{k+1} + \mathbf{p}_q, \quad U_{sim_{k+1}} = \frac{\mathbf{hs}_{k+1} \odot \mathbf{p}_{s_{top}}}{\|\mathbf{hs}_{k+1}\| \|\mathbf{p}_{s_{top}}\|}, \end{aligned} \quad (6)$$

where $\mathbf{f}_L(x, [\mathbf{hs}, \mathbf{cs}])$ is a standard LSTM cell with input x , $\mathbf{hs}$ and $\mathbf{cs}$ are the hidden state and the cell state, respectively. q refers to each triple in the query set, while s_{top} denotes the triple with the highest confidence in the support set (if $|\mathcal{S}_r| > 1$). k represents the k -th recurrent step.

Here, we deviate from the conventional approach of averaging representations in the support set and instead select the triple with the highest confidence as the reference for similarity calculation. This is because our goal is to capture the uncertainty of the triples through the matching processor. If a triple is completely credible, then the confidence of this triple completely similar to it should be 1. However, not all triples in the support set can be considered completely credible. After obtaining the uncertainty similarity scores, we utilize a confidence learner to predict the confidence of query triples.

2) *Confidence Prediction for Triples in the Query Set*: After obtaining plausibility scores of query triples from RMM, along with the uncertainty similarity between the support and query triples, the confidence learner aims to predict the confidence of the query triples. In the implementation, the intuitive logistic function is utilized by the confidence learner,

$$C_q = \sigma(w(E_q * U_{sim}) + b), \quad (7)$$

where σ is the sigmoid activation function, w and b are the learnable weight and bias, respectively. E is the plausibility score of the query triple computed by (2). And U_{sim} is the uncertainty similarity between the query triple and the most credible triple in the support set computed by (6). C_q is the predicted confidence for the query triple. With the confidence learner, MMUC fuses plausibility scores and metric-based similarity scores to generate more reliable confidence predictions. This hybrid design enables MMUC to deliver more calibrated and fine-grained uncertainty estimations even in data-scarce environments.

D. Training Objective

The training loss of MMUC consists of two parts: margin-based ranking loss (L_1) and MSE loss (L_2). For rank loss, we input triples in the query set of all training tasks to (3). And for MSE loss, we use the confidence learner to obtain predicted confidence of triples in the query set of all training tasks, then

Algorithm 1: Training Procedure of MMUC.

Require: Training task distribution $p(\mathcal{T})$
Ensure: Model parameters Θ

- 1: Initialize Θ
- 2: **while** early stopping criterion not met **do**
- 3: Sample a batch of tasks $\mathcal{B} \sim p(\mathcal{T})$
- 4: **for** each task $\mathcal{T}_r = (\mathcal{S}_r, \mathcal{Q}_r) \in \mathcal{B}$ **do**
- 5: $\mathbf{R}_{\mathcal{T}_r} \leftarrow \text{MLP}(\mathcal{S}_r; \Theta) \triangleright (1)$
- 6: $\mathbf{R}'_{\mathcal{T}_r} \leftarrow \mathbf{R}_{\mathcal{T}_r} - \delta \cdot \nabla_{\mathbf{R}} \mathcal{L}_{LP}(\mathcal{S}_r) \triangleright (4)$
- 7: **for** each query $(h_q, t_q) \in \mathcal{Q}_r$ **do**
- 8: $E_q \leftarrow \|\mathbf{h}_q + \mathbf{R}'_{\mathcal{T}_r} - \mathbf{t}_q\| \triangleright (2)$
- 9: $\mathbf{p}_s, \mathbf{p}_q \leftarrow \text{TripleRep}(\mathcal{S}_r, (h_q, t_q); \mathbf{R}'_{\mathcal{T}_r}) \triangleright (5)$
- 10: $U_{sim} \leftarrow \text{Match}(\mathbf{p}_s, \mathbf{p}_q) \triangleright (6)$
- 11: $C_q \leftarrow \sigma(w \cdot (E_q \odot U_{sim}) + b) \triangleright (7)$
- 12: Update $\mathcal{L}_{\text{link}}, \mathcal{L}_{\text{conf}}$ with E_q, C_q
- 13: **end for**
- 14: **end for**
- 15: $\mathcal{L} \leftarrow \alpha \cdot \mathcal{L}_{\text{link}} + \beta \cdot \mathcal{L}_{\text{conf}} \triangleright (9)$
- 16: Gradient update
- 17: **end while**
- 18: **return** Θ

calculate as follows:

$$\mathcal{L}_1 = \sum_r \mathcal{L}_{LP}(\mathcal{Q}_r),$$

$$\mathcal{L}_2 = \sum_r \sum_{((h,r,t),c) \in \mathcal{Q}_r} |C - c|^2. \quad (8)$$

Finally, we define the objective function of MMUC as

$$\mathcal{L} = \alpha \mathcal{L}_1 + \beta \mathcal{L}_2, \quad (9)$$

where α and β are trade-off factors. During training, our objective is to minimize joint loss $\mathcal{L}$.

The workflow of MMUC is summarized in Algorithm 1.

IV. EXPERIMENT SETTINGS

A. Datasets

Zhang et al. [17] first formalized the problem of few-shot UKGC and constructed datasets with varying levels of noisy triples (i.e., uncertainty levels) based on NL27K [6] to simulate real-world UKGs built with less human supervision through automated processes. Following their approach, we use NL27K-N0, NL27K-N1, NL27K-N2, and NL27K-N3 for evaluation, which consist of 0%, 10%, 20%, and 40% noisy triples respectively. We follow GMUC and use 101/13/20 tasks for training/validation/testing separately. Additionally, we constructed few-shot tasks based on CN15K [6] to further validate MMUC's performance. In CN15 K, relations with less than 1000 but more than 50 triples are kept, and we use 13/1/3 tasks for training/validation/testing. The statistics on the NL27 K and CN15 K datasets are shown in Table I. A detailed description of the dataset construction is provided in Section 3.1 of the supplemental material.

TABLE I
STATISTICS OF THE EXPERIMENTAL DATASETS

Dataset	Entities	Relations	Triples	Tasks	Avg(c)	Std(c)
NL27K	27,221	404	175,412	134	0.797	0.242
CN15K	15,000	36	241,158	17	0.629	0.232

B. Evaluation Protocol

Two types of metrics across all datasets are calculated to evaluate the overall performance of our framework and the baselines for the few-shot UKGC. For link prediction, we employ MRR and Hits@N. MRR is the Mean Reciprocal Rank of the correct entities, and Hits@N is the proportion of correct entities ranked in the top N. In our experiments, N is set to 1 and 10. Both of these metrics follow the principle of "higher values indicate better performance". For confidence prediction, we employ two common metrics in regression problems: Mean Squared Error (MSE) and Mean Absolute Error (MAE). In both cases, a lower value indicates better performance.

C. Implementation Details

All experiments were implemented using PyTorch on GeForce GTX 2080Ti GPUs. Due to space limitations, comprehensive hyperparameter settings and detailed model configurations are provided in Section 3.3 of the SM file.

V. EXPERIMENT RESULTS

A. Link Prediction

The objective of link prediction is for the model to accurately predict the tail entity based on the provided head entity and relation. In few-shot link prediction, during the testing phase, given a task about relation r including support triples, the model is expected to predict the tail entity, with the head entity provided in query triples. We adopt plausibility scores predicted by RMM as the ranking criterion for tail entity prediction. Moreover, we employ the same filtering strategy as baselines, where the model eliminates all correct tail entities except the ones to be predicted while sorting all candidates. The link prediction results on NL27 K and CN15 K datasets are shown in Tables II and III.

Among the baselines, GMatching [16] and FSRL [21] are classic few-shot link prediction methods designed for DGKs, both based on metric learning. MetaR [22], also for few-shot DGKs, is a representative meta-learning-based method for link prediction. NP-FKGC [26] is based on neural processes, and SIM [27] adopts a Transformer-based approach. It is important to note that these models are specifically designed for few-shot KGC and do not support confidence prediction, as they lack modules and loss functions tailored for uncertainty modeling. As a result, these models focus solely on optimizing link prediction using relatively simpler loss structures, without addressing the challenge of uncertainty in few-shot UKGs. UKGE [6] is the representative work in UKGC, capable of handling uncertainty, but it is not specifically designed for the few-shot setting. GMUC [17] is, to the best of our knowledge, currently the leading model for few-shot UKGC, which can simultaneously

TABLE II
EVALUATION RESULTS OF LINK PREDICTION ON NL27 K DATASETS. BOLD NUMBERS INDICATE THE BEST RESULTS.

Model	NL27K-N0			NL27K-N1			NL27K-N2			NL27K-N3		
	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10
GMatching	0.359	0.266	0.531	0.185	0.114	0.314	0.121	0.061	0.242	0.028	0.009	0.076
	± 0.010	± 0.012	± 0.008	± 0.009	± 0.011	± 0.011	± 0.007	± 0.005	± 0.011	± 0.011	± 0.009	± 0.013
MetaR	0.473	0.369	0.661	0.234	0.156	0.371	0.193	0.126	0.320	0.150	0.099	0.236
	± 0.022	± 0.014	± 0.010	± 0.007	± 0.006	± 0.005	± 0.010	± 0.014	± 0.003	± 0.005	± 0.007	± 0.007
UKGE	0.053	0.044	0.130	0.185	0.071	0.107	0.056	0.066	0.143	0.092	0.088	0.140
	± 0.003	± 0.009	± 0.005	± 0.009	± 0.002	± 0.001	± 0.001	± 0.003	± 0.006	± 0.003	± 0.003	± 0.004
FSRL	0.314	0.240	0.477	0.201	0.120	0.367	0.179	0.107	0.342	0.118	0.054	0.228
	± 0.009	± 0.009	± 0.013	± 0.006	± 0.007	± 0.008	± 0.003	± 0.003	± 0.005	± 0.016	± 0.021	± 0.015
NP-FKGC	0.388	0.204	0.685	0.156	0.031	0.375	0.053	0.001	0.196	0.101	0.041	0.197
	± 0.005	± 0.006	± 0.007	± 0.003	± 0.001	± 0.005	± 0.002	± 0.001	± 0.010	± 0.002	± 0.002	± 0.004
SIM	0.183	0.162	0.209	0.163	0.151	0.176	0.137	0.119	0.165	0.165	0.154	0.179
	± 0.014	± 0.015	< 0.001	± 0.025	± 0.020	± 0.036	± 0.016	± 0.025	± 0.010	± 0.007	± 0.005	± 0.010
GMUC	0.428	0.328	0.621	0.201	0.131	0.343	0.136	0.103	0.288	0.141	0.098	0.193
	± 0.005	± 0.007	± 0.012	± 0.010	± 0.012	± 0.008	± 0.007	± 0.007	± 0.016	± 0.008	± 0.008	± 0.010
Meta-KGR	0.150	0.087	0.262	0.139	0.094	0.221	0.112	0.062	0.201	0.090	0.043	0.174
	± 0.151	± 0.126	± 0.222	± 0.150	± 0.133	± 0.200	± 0.132	± 0.112	± 0.191	± 0.100	± 0.059	± 0.180
MMUC	0.508	0.402	0.702	0.306	0.226	0.452	0.230	0.163	0.357	0.171	0.114	0.272
	± 0.009	± 0.010	± 0.011	± 0.011	± 0.012	± 0.010	± 0.006	± 0.007	± 0.008	± 0.010	± 0.012	± 0.018

Each result is reported as mean $\pm$ standard deviation.

TABLE III
EVALUATION RESULTS OF LINK PREDICTION ON CN15 K. BOLD NUMBERS ARE THE BEST RESULTS.

Model	MRR	Hits@1	Hits@10
GMatching	0.081 $\pm$ 0.003	0.016 $\pm$ 0.006	0.160 $\pm$ 0.003
MetaR	0.096 $\pm$ 0.002	0.037 $\pm$ 0.002	0.222 $\pm$ 0.013
UKGE	0.022 $\pm$ 0.002	0.007 $\pm$ 0.002	0.049 $\pm$ 0.003
FSRL	0.094 $\pm$ 0.003	0.040 $\pm$ 0.003	0.186 $\pm$ 0.002
NP-FKGC	0.133$\pm$0.016	0.013 $\pm$ 0.009	0.356$\pm$0.070
SIM	0.078 $\pm$ 0.003	0.032 $\pm$ 0.005	0.122 $<$ 0.001
GMUC	0.101 $\pm$ 0.002	0.036 $\pm$ 0.003	0.215 $\pm$ 0.011
Meta-KGR	0.015 $\pm$ 0.015	0.004 $\pm$ 0.005	0.041 $\pm$ 0.039
MMUC	0.107 $\pm$ 0.006	0.043$\pm$0.010	0.247 $\pm$ 0.003

Each result is reported as mean $\pm$ standard deviation.

perform link prediction and confidence prediction in few-shot context.

As seen in Table II, even under this advantageous setup for KGC-only baselines, MMUC achieves near-best or best results on all NL27 K variants (N0–N3) for link prediction metrics. This demonstrates that MMUC’s joint optimization of link prediction and confidence regression does not compromise its ranking ability. Instead, the integration of relation-centric representation and confidence-guided learning leads to enhanced overall predictive power. For instance, MMUC attains an MRR of 0.508 and Hits@10 of 0.702 on NL27K-N0, outperforming all baselines by a clear margin. Moreover, MMUC’s performance remains stable even in few-shot datasets that contain noise. On the high-noise few-shot dataset NL27K-N3 (40% noisy samples), accurately predicting the tail entity in the top ten rank is a challenging task. An exception worth noting is SIM, which achieves the highest Hits@1 score of 0.154 on NL27K-N3. However, it does not maintain this advantage in Hits@10, where MMUC surpasses it by a significant margin (0.272 vs. 0.179). This discrepancy between Hits metrics suggests that while SIM may be good at capturing high-confidence top-1 predictions for few triples, it lacks consistent generalization across the candidate space,

especially in sparse or noisy environments. In contrast, MMUC provides more stable and broadly accurate rankings under varying noise conditions, indicating it is capable of filtering out noise interference to some extent and quickly capturing semantic information in UKGs.

Another noteworthy model is NP-FKGC, whose performance exhibits unexpected variability across datasets. Architecturally, NP-FKGC adopts a more complex design compared to other baselines. For instance, on the CN15 K dataset, the total parameter count of NP reaches 24,607,211, which is substantially higher than MMUC’s 14,209,220 parameters. Despite its size and complexity, NP-FKGC’s link prediction performance varies significantly across datasets. It achieves relatively strong results on CN15 K, a dataset with only 36 relations. However, its Hits@1 score remains notably low, indicating potential challenges in making precise top-1 predictions even in simpler settings. In contrast, on the more relation-rich NL27 K series, where the number of relations and structural complexity increase, NP-FKGC fails to maintain competitive performance. For instance, its MRR drops to 0.156 on NL27K-N1 and 0.053 on NL27K-N2, both of which are significantly lower than MMUC and even simpler baselines like MetaR. This drop suggests that NP-FKGC’s performance does not scale well with increased task complexity or sparsity, possibly due to its high model variance and overfitting tendencies under few-shot supervision. These findings further validate the robustness of MMUC. Despite having fewer parameters, MMUC generalizes better across both simple and complex datasets, and consistently achieves higher Hits@1 and MRR, illustrating its balanced design between expressiveness and generalization capacity under few-shot UKGC settings.

We also included Meta-KGR [36], a reinforcement learning-based method, to evaluate our framework against RL approaches. As shown in Tables II and III, MMUC consistently outperforms Meta-KGR across all datasets. For instance, on the NL27K-N0 dataset, MMUC achieves significantly higher MRR and Hit@10. This performance advantage can be attributed to the inherent structural challenges in few-shot UKGs.

TABLE IV
EVALUATION RESULTS OF CONFIDENCE PREDICTION ON NL27 K AND CN15 K DATASETS. BOLD NUMBERS ARE THE BEST RESULTS.

Model	NL27K-N0	NL27K-N1	NL27K-N2	NL27K-N3	CN15K
	MSE / MAE	MSE / MAE	MSE / MAE	MSE / MAE	MSE / MAE
UKGE	0.073±0.005 / 0.202±0.005	0.063±0.004 / 0.196±0.029	0.074±0.008 / 0.192±0.012	0.091±0.012 / 0.217±0.021	0.473±0.026 / 0.677±0.021
GMUC	0.012±0.001 / 0.084±0.001	0.019±0.001 / 0.097±0.002	0.024±0.001 / 0.104±0.002	0.027±0.001 / 0.115±0.004	0.022±0.002 / 0.110±0.002
MMUC	0.012±0.001 / 0.084±0.001	0.014±0.001 / 0.088±0.003	0.019±0.002 / 0.097±0.005	0.021±0.003 / 0.112±0.005	0.020±0.001 / 0.106±0.003

Each result is reported as mean ± standard deviation.

RL-based models like Meta-KGR typically rely on finding reliable reasoning paths in the graph to train the agent. However, in our setting, the graph is characterized by both neighbor sparsity and uncertainty, making it difficult for the RL agent to navigate and discover valid paths. In contrast, MMUC minimizes reliance on complex path structures by employing a relation-centric metric learning paradigm, thereby demonstrating superior robustness and generalization capability even when effective reasoning paths are scarce. In addition, we also provide experimental results and analysis of MMUC and baseline models on NEL-one and Wiki-One datasets (Section 4.4 of SM File).

UKGE and GMUC serve as baselines that can simultaneously perform both link prediction and confidence prediction, with a more complex joint loss that is harder to optimize. Among them, GMUC, specifically designed for few-shot scenarios, is the core baseline for comparison. From the results, our method consistently outperforms GMUC in various link prediction metrics, demonstrating its superior performance and robustness in few-shot UKGC.

B. Confidence Prediction

The objective of this task is to predict confidence scores of unseen relation facts in the query set after observing few samples in the support set, that is $((h, r, t), ?)$ where $(h, r, t) \in \mathcal{Q}_r$ and $\mathcal{Q}_r \in \mathcal{T}_{test}$ which means the relation r has never appeared in the training process. The results of confidence prediction on all datasets are shown in Table IV. It can be seen that MMUC outperforms all baselines consistently. Notice that the performance improvements of MMUC in NL27 K datasets (NL27K-N1/N2/N3), which are mixed with noise, are more obvious. This suggests that MMUC is particularly effective in handling noisy data, as it can effectively capture the underlying uncertain knowledge in KGs despite the presence of noise. Overall, the results demonstrate the effectiveness of MMUC in the context of few-shot learning even with noisy samples.

We conduct all of the above experiments multiple times, running each scenario with different random seeds to mitigate any potential bias due to the variability inherent in small sample sizes. Compared with GMUC, MMUC demonstrates significant statistical results across both link prediction (with P-values $\leq 5.29\text{E-}06$) and confidence prediction (with P-values $\leq 9.23\text{E-}04$). These results highlight MMUC's superior performance, and the statistical significance of the improvements confirms that these differences are not due to random chance.

These consistent gains in both tasks demonstrate that MMUC's dual-module architecture, which integrates relation meta-learning and confidence-aware metric learning, offers a more expressive and effective framework compared to GMUC's

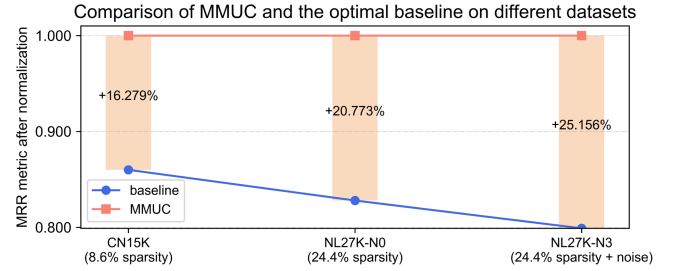


Fig. 4. Comparison of MMUC and the baseline GMUC on different datasets. The MRR metric, min-max normalized for comparison, shows MMUC (square points) and GMUC (round points) performances, with bars indicating the percentage improvement of MMUC over the baseline.

neighbor-aggregation-based design. While GMUC remains sensitive to neighbor sparsity because of its reliance on background graph structure, MMUC maintains its robustness by shifting the focus to relation-centric representations and support-query matching, successfully overcoming the limitations GMUC faces in sparse or noisy environments.

VI. DISCUSSION

A. Results in a Neighbor Sparsity View

MMUC demonstrates superior performance across all datasets, leading us to explore the connection between performance gains and dataset characteristics. We analyze the neighbor sparsity of entities in the background graphs of two benchmarks. Standardizing sparsity as entities with neighbors less than 1/10 of the average degree (4 for NL27 K, 8 for CN15 K), we found 8.6% sparse neighbors in CN15 K and 24.4% in NL27 K. Utilizing MRR as an example, we normalize metrics through min-max normalization across multiple experiments. The results, illustrated in Fig. 4, indicate MMUC's greater performance improvement in NL27 K, where neighbor sparsity is more pronounced. This suggests that MMUC's dual-module synergy effectively avoids the impact of neighbor sparsity. Moreover, MMUC's notable improvement in the highly noisy NL27K-N3 dataset highlights its robustness to noise, confirming the framework's ability to handle noise in uncertain KGs.

B. Robustness for Neighbor Sparsity

To demonstrate that MMUC is an effective solution to the challenge of neighbor sparsity, we simulated background graphs with varying sparsity levels using real-world uncertain knowledge graphs. As shown in Fig. 5, the performance of the baseline model degrades significantly as sparsity increases, while MMUC maintains stable results. This robustness stems from

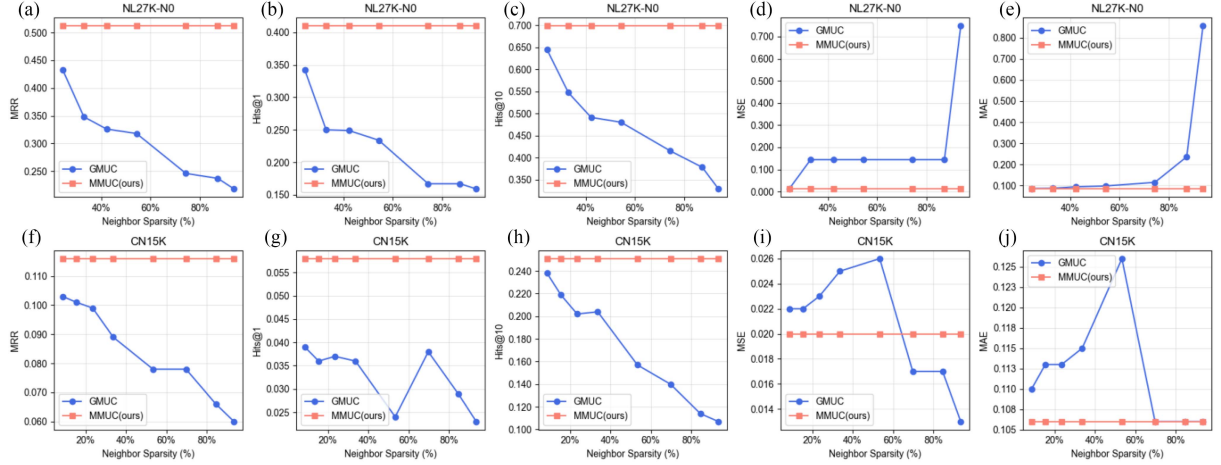


Fig. 5. Performance comparison of MMUC with the baseline GMUC on simulated graphs exhibiting varying degrees of neighbor sparsity from real-world UKGs.

TABLE V

RESULTS OF MMUC'S ABLATION STUDY ON LINK PREDICTION AND CONFIDENCE PREDICTION IN NL27K-N0 AND NL27K-N2. BOLD NUMBERS ARE THE BEST RESULTS.

Model	NL27K-N0					NL27K-N2				
	MRR↑	Hits@1↑	Hits@10↑	MSE↓	MAE↓	MRR↑	Hits@1↑	Hits@10↑	MSE↓	MAE↓
-UG	0.502±0.010 (p=0.019)	0.396±0.012 (p<<0.001)	0.696±0.008 (p=0.028)	0.014±0.002 (p=0.001)	0.087±0.005 (p=0.006)	0.221±0.007 (p=0.002)	0.154±0.009 (p=0.876)	0.336±0.011 (p<<0.001)	0.025±0.007 (p=0.001)	0.111±0.013 (p<<0.001)
-CMM	0.454±0.004 (p<<0.001)	0.350±0.009 (p<<0.001)	0.657±0.013 (p<<0.001)	0.016±0.001 (p<<0.001)	0.100±0.002 (p<<0.001)	0.158±0.008 (p<<0.001)	0.087±0.011 (p<<0.001)	0.265±0.009 (p<<0.001)	0.021±0.001 (p<<0.001)	0.107±0.002 (p<<0.001)
-STC	0.487±0.006 (p<<0.001)	0.384±0.007 (p<<0.001)	0.676±0.009 (p<<0.001)	0.016±0.004 (p=0.001)	0.089±0.009 (p=0.011)	0.202±0.011 (p<<0.001)	0.133±0.011 (p<<0.001)	0.324±0.013 (p<<0.001)	0.062±0.013 (p<<0.001)	0.171±0.015 (p<<0.001)
-TR	0.509±0.017 (p=0.317)	0.402±0.021 (p=0.054)	0.699±0.015 (p=0.212)	0.047±0.002 (p<<0.001)	0.155±0.002 (p<<0.001)	0.185±0.013 (p<<0.001)	0.118±0.012 (p<<0.001)	0.310±0.017 (p<<0.001)	0.021±0.001 (p<<0.001)	0.099±0.002 (p<<0.001)
-CL	0.497±0.013 (p<<0.001)	0.391±0.014 (p<<0.001)	0.694±0.015 (p=0.023)	0.057±0.010 (p<<0.001)	0.166±0.011 (p<<0.001)	0.203±0.005 (p<<0.001)	0.139±0.006 (p<<0.001)	0.311±0.005 (p<<0.001)	0.027±0.001 (p<<0.001)	0.111±0.002 (p<<0.001)
MMUC	0.511±0.008	0.410±0.007	0.702±0.011	0.012±0.001	0.084±0.001	0.229±0.009	0.151±0.007	0.379±0.023	0.019±0.002	0.097±0.005

Each result is reported as mean ± standard deviation. Paired T-tests are performed against MMUC, and P-values are shown in parentheses.

MMUC's dual-module architecture, which bypasses entity-centric neighbor aggregation and instead learns from relation-centric training task-local information.

From a theoretical perspective, this design decouples both relation meta representation learning (via RMM) and uncertainty similarity matching (via CMM) from the entity's local neighborhood. As both modules operate solely on support triples and query triples from training tasks rather than background structures, MMUC remains unaffected even under extreme sparsity, as validated by our simulations. For more detailed discussion and supporting analysis, please refer to Section 2.1 of the supplemental material.

C. Ablation Study

MMUC adopts a synergistic approach, and we aim to assess the individual and collective contributions of these strategies to its overall performance. We conduct the ablation study, performing multiple runs for each model variant to compute the mean and standard deviation, and further apply paired T-tests to assess the statistical significance of performance differences compared to our MMUC model, with the results shown in Table V. Limited in paper length, only NL27K-N0 and NL27K-N2 representing datasets without and with noise are shown. The results for

additional datasets and analyses can be found in Section 4.3 of the supplemental material.

In the experiment, the P-value quantified the significance of the effect of module removal on the model performance through the T-test [37], providing a key basis for the statistical verification of the functions of each module. When $p \ll 0.001$, it indicates that the performance degradation caused by module absence is highly unlikely to be caused by random errors. If p is between 0.01 and 0.05, it indicates that the module has a substantial impact on specific indicators. When $p \geq 0.1$, it means that the impact of module missing on performance is not significant. From the results, P-values of the performance changes of most modules after removal are less than 0.05, indicating that these modules have a significant impact on the model performance.

Note that RMM is a crucial component of MMUC and completely removing it would render it impossible to address the few-shot uncertain KGC. In essence, RMM is an uncertainty generalized few-shot meta-learning module, providing the plausibility scores of triples. Specifically, we explore the effects of removing the uncertain generalized part in this module and replacing it with the conventional margin rank loss, which does not consider uncertainty (-Uncertain Generalization, -UG). Additionally, we assess the performance by excluding

the entire CMM and utilizing RMM for link prediction. In this case, we introduce the logistic function to fit plausibility scores to confidence values for the purpose of confidence prediction (**-Confidence Metric Module, -CMM**). Both ablations resulted in a decline in performance, with the removal of the uncertain generalized part having a serious impact on confidence prediction. This is because the conventional rank loss only considers semantic information and ignores confidence, leading to the disregard of uncertainty in both plausibility and relational meta-information used in CMM. Removing CMM and directly mapping plausibility to confidence not only reduces the accuracy of confidence prediction but also leads to a decline in link prediction performance. While RMM is responsible for capturing the semantics relation facts with uncertainty, CMM complements this and provides a comprehensive understanding of the uncertainty of facts. By working together, these two modules enable MMUC to accurately capture the semantic information with uncertainty in few-shot UKGs.

Then, we perform ablation experiments on two strategies in CMM. One potential improvement we made is to adjust the support sample strategy in the uncertainty similarity matching processor. Rather than selecting the triple with the highest confidence, we take the average of all three triple representations here (**-Support Top Confidence, -STC**). The resulting performance drop suggests that selecting the most reliable support triple is crucial for maintaining robustness in uncertain and few-shot scenarios. We also use entity pair representations as inputs in the uncertainty similarity matching processor to assess whether the triple representation is effective (**-Triple Representation, -TR**). The results confirm the effectiveness of integrating relation representations, as the inclusion of relations can enhance the semantics of facts.

Besides, we evaluate the impact of the confidence learner. Specifically, we use similarities from the matching processor directly through the logistic function as predicted results (**-Confidence Learner, -CL**), which is equivalent to using the embedding representation from RMM without considering plausibility scores of triples. This is a simple superposition of meta and metric matching without considering the potential interaction. The experimental results show that this leads to a noticeable decline, particularly under noisy conditions. This confirms the necessity of modeling the non-linear interaction between semantic plausibility and uncertainty-aware similarity.

The combined impact of all components, as evidenced by performance improvements across nearly all metrics and datasets, is further supported by rigorous multi-run experiments and paired T-tests. These statistical evaluations confirm that the observed performance gains are not incidental but stem from the deliberate and complementary design of each module within MMUC. This comprehensive validation highlights the robustness and effectiveness of MMUC's architecture in jointly addressing few-shot and uncertainty-aware knowledge graph completion.

D. Case Study of Sparse and Dense Entities

To further demonstrate the practical effectiveness of MMUC in addressing neighbor sparsity, we conduct a detailed case study

comparing MMUC with the baseline GMUC on representative entities with varying neighborhood densities. The selected entities are categorized as either sparse (e.g., clean clothes, boredom, play games, socialize) or dense (e.g., cat, child, cook, person) based on their number of neighbors in the background graph.

As illustrated in Fig. 6, MMUC consistently achieves higher MRR and Hits@10 scores than GMUC across all evaluated entities, regardless of their neighborhood density. Notably, for highly sparse entities like "clean clothes," MMUC attains an MRR of 0.506 and Hits@10 of 0.500, far exceeding the results of GMUC and demonstrating robust predictive power even with minimal neighborhood information. This trend persists across other sparse cases, such as "play games" and "socialize," as well as for dense entities, where MMUC continues to provide competitive or superior results.

It is also important to highlight that entities with sparse neighborhoods often have limited appearances in the training data, making them inherently challenging for representation learning. The strong performance of MMUC on these difficult cases underscores the effectiveness of our approach in overcoming the limitations imposed by neighbor sparsity. These case-based findings provide concrete evidence of MMUC's robustness and further validate its advantage over conventional metric learning methods, which tend to rely heavily on dense neighborhood information.

Furthermore, to strictly validate the model's resilience under limit conditions, we extended our evaluation to extreme sparsity scenarios (e.g., where background graph deletion rates reach 80% and 90%). The detailed experimental results and the specific correspondence between deletion rates and entity sparsity are provided in Fig. 6 and Section 4.5 of the SM file. These additional experiments confirm that MMUC maintains stable and superior performance compared to baselines even when the background structure is severely compromised, further verifying the effectiveness of our relation-centric meta-learning strategy.

Complementing the metric comparison in Fig. 6, we further provide a concrete look at the prediction rankings in Table VI. We selected two representative queries: a sparse case (city:panaji, city located in geopolitical location, ?) and a dense case (insect:insects, animal such as invertebrate, ?).

As shown in the table, for the sparse query, MMUC accurately identifies the ground truth state or province:goa at **Rank 1**, whereas the baseline GMUC drops it to Rank 10, likely due to the lack of neighborhood structural information. Similarly, in the dense scenario, MMUC maintains its superiority by ranking the target insect:gnats at **Rank 1**, compared to GMUC's Rank 18. These specific examples intuitively validate MMUC's robustness against neighbor sparsity.

VII. CONCLUSION

In this paper, we propose a meta-metric framework (MMUC) for few-shot uncertain knowledge graph completion. To fundamentally address the challenge of neighbor sparsity, MMUC introduces a paradigm shift from entity-centric to relation-centric

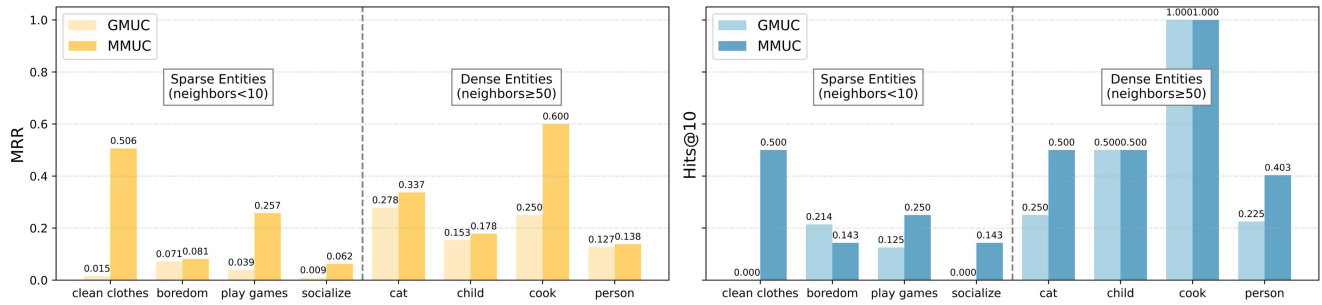


Fig. 6. Comparison of MMUC and GMUC performance on entities with sparse (fewer than 10 neighbors) and dense (50 or more neighbors) neighborhoods from CN15 K. Results for both MRR (left) and Hits@10 (right) show that MMUC consistently outperforms GMUC, particularly for entities with severe neighbor sparsity.

TABLE VI
CASE STUDY OF LINK PREDICTION RESULTS FOR REPRESENTATIVE SPARSE AND DENSE ENTITIES

Rank	Sparse		Dense	
	MMUC	GMUC	MMUC	GMUC
1	stateorprovince:goa	city:division	insect:insects	insect:insects
2	sportsteam:uconn	city:philadelphia	insect:gnats	insect:beneficial_insects
3	city:vegas	country:iraq	insect:termite	animal:organisms
4	country:pakistan	city:orlando	animal:scavengers	crustacean:topic
5	sportsteam:pittsburgh_pirates	organization:australia	animal:creatures	insect:mosquitoes
6	island:ios_apps	city:san_francisco	animal:carnivores	insect:bees
7	city:amritsar	city:amsterdam	invertebrate:stages	insect:insect_pests
8	island:west_point	sportsteam:pittsburgh_pirates	animal:organisms	insect:bugs
9	island:cayenne	sportsteam:buckeyes	animal:chicken001	insect:pests
10	island:ocean_springs	stateorprovince:goa	insect:parasitoids	arthropod:spiders
	GT Rank: 1	GT Rank: 10	GT Rank: 1	GT Rank: 18

learning. It achieves this by unifying relation-aware meta-learning and confidence-based metric learning into a synergistic framework. MMUC achieves superior results over baselines and demonstrates strong robustness in low-resource and uncertain contexts.

Nevertheless, MMUC still has certain limitations. First, similar to many existing few-shot KGC models, MMUC may encounter challenges with complex relation types such as 1-N, N-1, and N-N mappings. Our current architecture does not explicitly model the effect of these complex relations on uncertainty estimation, which may limit its applicability in knowledge graphs with highly diverse relational patterns [23]. Second, the current multi-task objective of MMUC uses a simple weighted sum of ranking loss and confidence regression loss, providing limited flexibility in dynamically balancing link prediction and confidence estimation [38]. Exploring more adaptive or task-aware loss functions—such as uncertainty weighting or multi-objective optimization—remains an important direction for future work.

Furthermore, future work can explore enhancing and extending MMUC along two main directions: 1) Framework Enhancement: Integrating advanced representation paradigms, such as multi-view clustering [39] or contrastive learning [40], could further improve the robustness and discriminative power of relation representations. 2) Practical Extensions: Adapting MMUC to privacy-sensitive federated learning scenarios [41] or

employing advanced knowledge transfer techniques [42] could facilitate its deployment in real-world, distributed environments.

We believe our work not only offers a practical and robust solution for few-shot uncertain KGC, but also provides new insights into relation-centric representation learning and uncertainty modeling. In future work, we will further investigate these issues to enhance the predictive capability of MMUC in complex scenarios.

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